Let the mind manage the body
Que l'esprit gère le corps

Index Number: .....

NATIONAL CERTIFICATE OF EDUCATION

2025

MATHEMATICS (N510)

TIME: 2 HOURS

Candidates answer on the Question Paper.
Additional Materials: Geometrical Instruments

READ THESE INSTRUCTIONS FIRST

1. Write your index number in the space provided above.
2. Write in dark blue or black ink.
3. You may use an HB pencil for diagrams. Do not use correction fluid.
4. Diagrams are not drawn to scale unless otherwise specified.
5. Answer **ALL** questions.
6. All workings should be shown in the spaces provided.
7. This document consists of **31** questions printed on **23** pages, numbered **2** to **24.**
8. Calculators must **NOT** be used for this paper.
9. The number of marks is given in brackets [ ] at the end of each question or part question.
10. The total number of marks for this paper is **100**.

For Examiners' use														
Page No.	**3**	**5**	**7**	**9**	**11**	**13**	**15**	**17**	**19**	**21**	**23**	**24**	**Total**	**Signature**
Examiner														
Team Leader														
CE/ACE														

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Marks

1. Work out:

$$\begin{array}{r} 8\ 4\ 9 \\ -\ 5\ 2\ 3 \\ \hline \\ \hline \end{array}$$

Answer: ..... [1]

2. Work out:

(a) $2.1 + 0.5$

Answer: ..... [1]

(b) $\frac{9}{17} - \frac{3}{17}$

Answer: ..... [1]

3. Find $\sqrt[3]{8}$

Answer: ..... [1]

4. State the **order** of the matrix.

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Answer: ..... [1]

5. Express 13% as a **fraction**.

Answer: ..... [1]

Marks

6. Circle the **composite** number in the list below.

2 13 27 41

[1]

7. Convert 5 L into mL.

Answer: ..... mL [1]

8. Evaluate $3 \times (4 - 2) + 5$

Answer: ..... [1]

9. (a) Expand $4(x + 2)$

Answer: ..... [1]

(b) Factorise $6x - 15$

Answer: ..... [1]

10. Simplify $(a^5)^4$

Answer: ..... [1]

E
TL
CE

Marks

11. **Circle the correct answer. Each item carries 1 mark.**

(a) What is the value of **3** in 21.03?

A 3 tens

B 3 tenths

C 3 hundreds

D 3 hundredths

(b) Which of the following is an **irrational** number?

A $\sqrt{25}$

B $\sqrt{5}$

C $\frac{22}{7}$

D 0.25

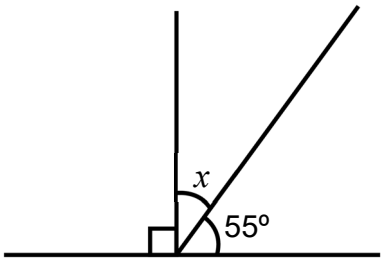
(c) What is the **value** of angle x?

A 35º

B 45º

C 125º

D 135º



(d) Convert 2 kg 350 g into grams.

A 2350 g

B 235 g

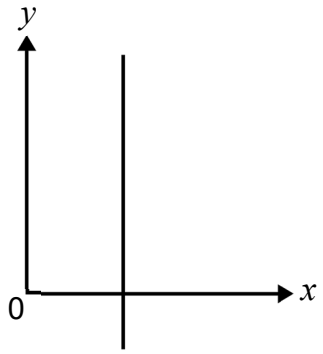
C 23.50 g

D 2.350 g

Marks

(e) Which one of the following lines has a **gradient** of zero?

A



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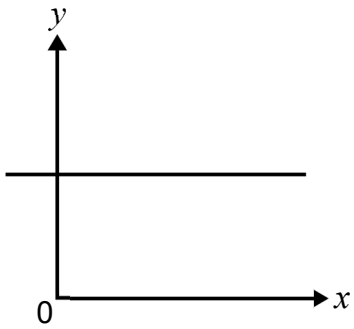

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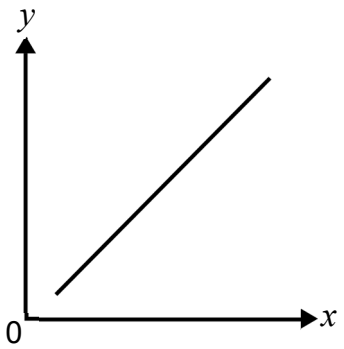

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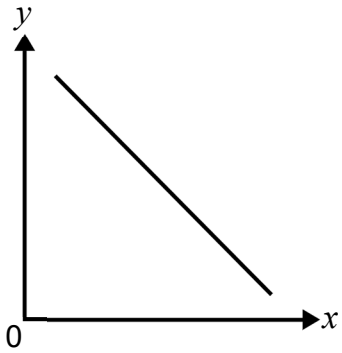


(f) A bag of flour weighs 15 kg. What is the mass of 4 such bags?

A 11 kg

B 19 kg

C 30 kg

D 60 kg

(g) What is the **L.C.M.** of $4x$ and $12y$?

A $4x$

B $12y$

C $12xy$

D $24xy$

E

TL

CE

Marks

12. A list of numbers is given below.

Circle the **smallest** number.

2.05 2.5 2.55 2.005

[1]

13. (a) Evaluate $\frac{1}{2} + \frac{1}{3}$

Answer: ..... [2]

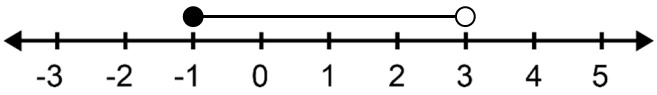
(b) Mira has Rs 360. She gives $\frac{2}{5}$ of her money to Tim.

How much money does Tim receive?

Answer: Rs ..... [2]

Marks

14. (a) Which inequality correctly describes the set of numbers shown on the number line?



Tick (✓) the correct inequality.

$-1 \leq x \leq 3$ ☐ $-1 < x \leq 3$ ☐

$-1 \leq x < 3$ ☐ $-1 < x < 3$ ☐

[1]

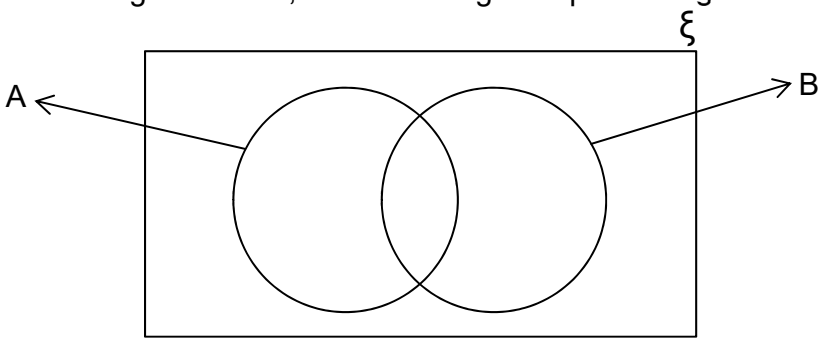
(b) Solve the inequality $-5x > 20$

Answer: …….. [2]

15. (a) Write down the missing term in the following sequence:

(2, 3), (3, 5), (4, 7), (……., …….) [1]

(b) In the Venn diagram below, shade the region representing A ∪ B.



[1]

E

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Marks

16. $\mathbf{P} = \begin{pmatrix} 8 \\ -5 \end{pmatrix}$ $\mathbf{Q} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

Find

(a) $\mathbf{P} - \mathbf{Q}$

Answer: …… [2]

(b) $2\mathbf{Q}$

Answer: …… [2]

17. (a) Find the **value** of m, if $27 = 3^m$.

Answer: $m =$ …….. [1]

(b) Hence, solve $3^{6x} \div 3^{4x} = 27$.

Answer: $x =$ …… [2]

Marks

18. A line passes through the points **A** (0, 1) and **B** (3, 7).

(a) Find the **gradient** of the line **AB**.

Answer: ..... [2]

(b) **State** the equation of the line **AB**.

Answer: ..... [1]

19. 12 men take 20 days to build a house.

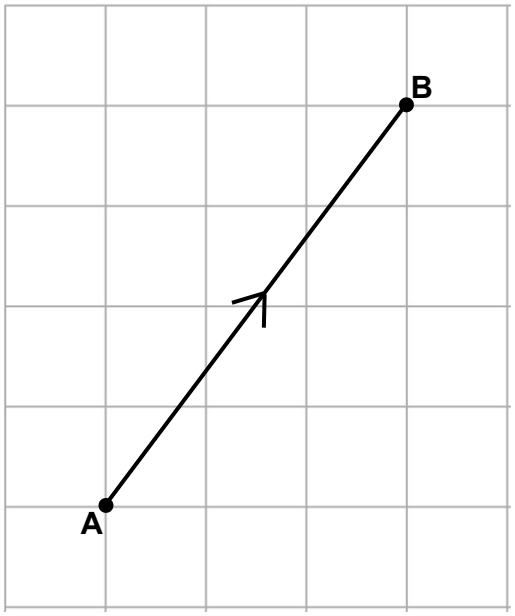
How many days would it take for 10 men to build a similar house?
[Assume that the men work at the same rate.]

Answer: ..... days [3]

E
TL
CE

Marks

20. Point **A** is mapped onto point **B** under a translation **T**.



(a) Tick (✓) the box, which correctly indicates **T** expressed as a column vector.

$\begin{pmatrix} -3 \\ -4 \end{pmatrix}$ ☐

$\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ ☐

$\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ ☐

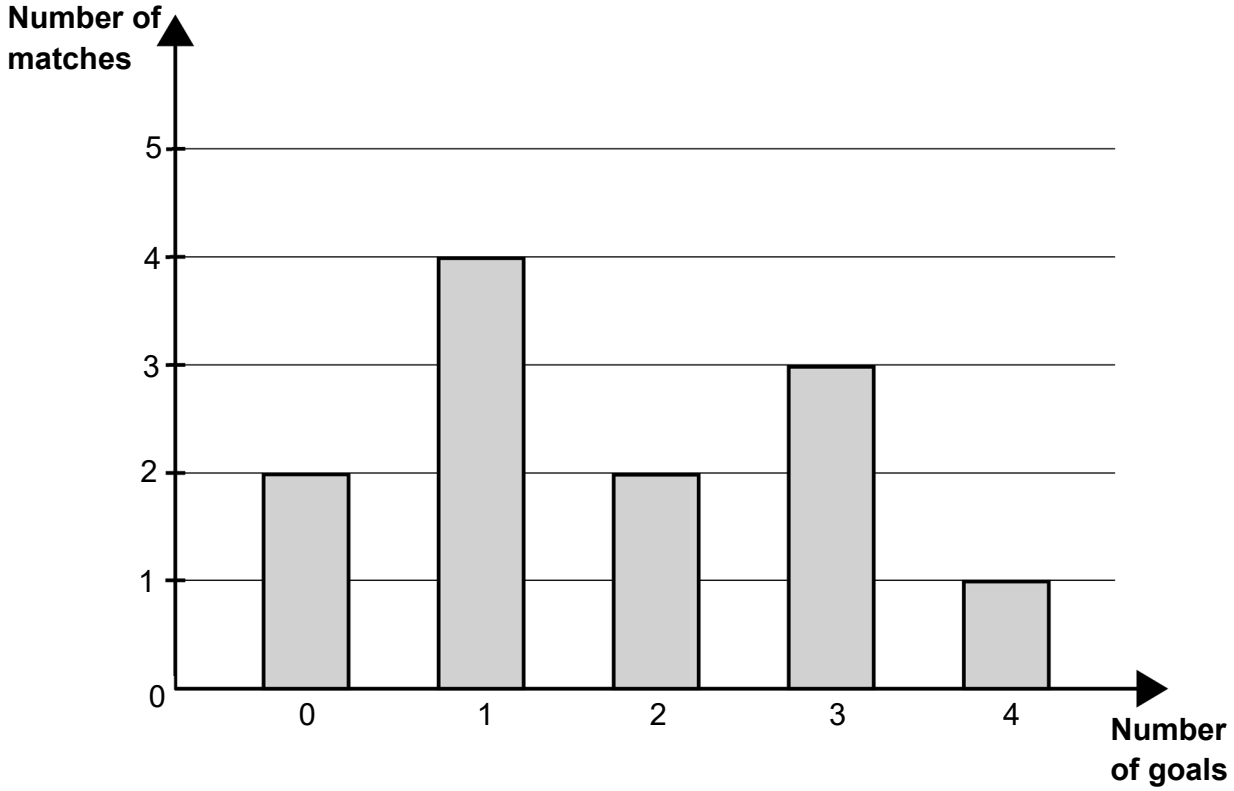
[1]

(b) Find the **magnitude** of the column vector **T**.

Answer: …… [2]

Marks

21. The bar chart shows the number of goals scored in 12 matches of a football tournament.



(a) Using the information given in the bar chart, complete the table.

Number of goals	0	1	2	3	4
Number of matches	.....	4	.....	.....	1

[2]

(b) Find the **mean** number of goals scored per match.

Answer: ..... goals per match [2]

E

TL

CE

Marks

22. (a) The plan of a house is drawn to a scale 1 : 400.
On the plan, a wall is represented by a line of 7.5 cm.

Find the actual length of the wall, in **metres**.

Answer: ..... m [2]

(b) Solve $p^2 - 10p + 10 = -14$.

Answer: ..... [3]

Marks

23. A formula is given by $A = \pi r^2 + \pi r l$.

(a) Calculate the value of A when $r = 7$ cm and $l = 4$ cm.

Leave your answer in terms of π.

Answer: A = ... [2]

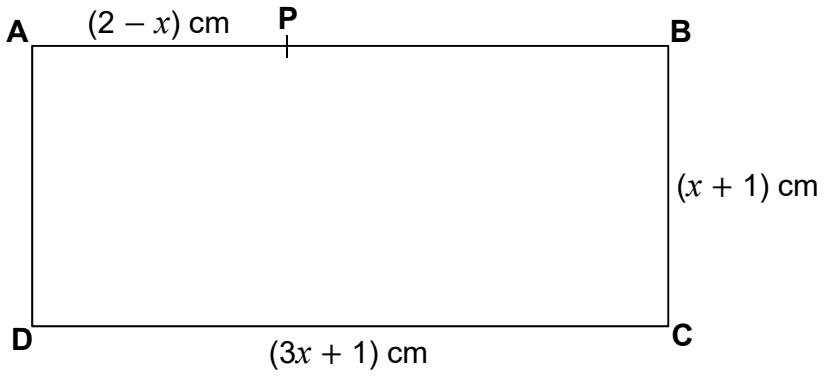
(b) Make l the subject of the formula, leaving your answer in terms of π.

Answer: ..... [2]

E
TL
CE

Marks

24. **ABCD** is a rectangle.
It is given that **DC** $= (3x + 1)$ cm and **BC** $= (x + 1)$ cm.
P is a point on **AB**, such that **AP** $= (2 - x)$ cm.



Giving each answer in its **simplest form**, find, in terms of x, an expression for

(a) **PB**,

Answer: ..... cm [2]

(b) area of rectangle **ABCD**.

Answer: ..... cm^2 [2]

Marks

25. Solve the simultaneous equations:

$$3x - y = 7$$

$$2x + 3y = 1$$

Answer: $x =$ **……**, $y =$ **……** [4]

E
TL
CE

Marks

26. The exterior angles of a triangle are in the ratio 5 : 6 : 7.

Calculate

(a) the difference between the greatest and the smallest **exterior** angles,

Answer: ……° [4]

(b) the greatest **interior** angle of the triangle.

Answer: ……° [2]

Marks

27. John deposits a sum of Rs 200 000 for a period of 30 months in a bank.
The rate of simple interest is 3% per annum.

Find the **total** amount of money that John will get from the bank when he withdraws all his money after 30 months.

Answer: Rs …….. [3]

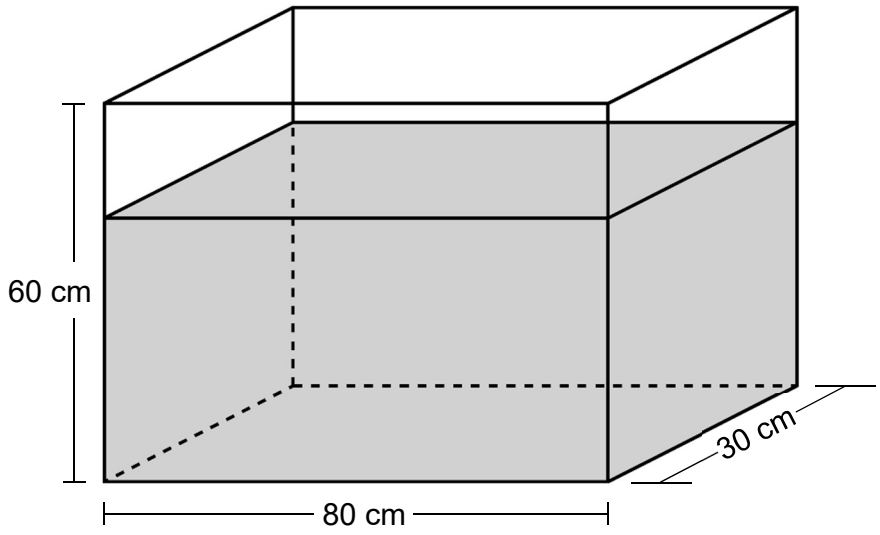
E

TL

CE

Marks

28. The diagram shows a rectangular water tank **A** with dimensions 80 cm by 30 cm by 60 cm.



Tank **A**

(a) Tank **A** is $\frac{2}{3}$ full of water.
Calculate the volume of water in Tank **A**.

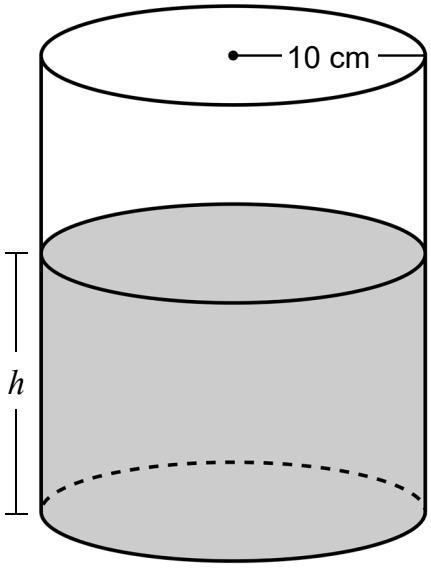
Answer: ..... cm^3 [3]

Marks

(b) The water from tank **A** is poured into a cylindrical container **B** with radius 10 cm.

While the water from tank **A** is being poured into container **B**, 6 L of water is **lost**.

Find the depth of the water, h cm, in container **B**. Give your answer in terms of π.



Container **B**

Answer: $h =$ ..... cm [4]

E

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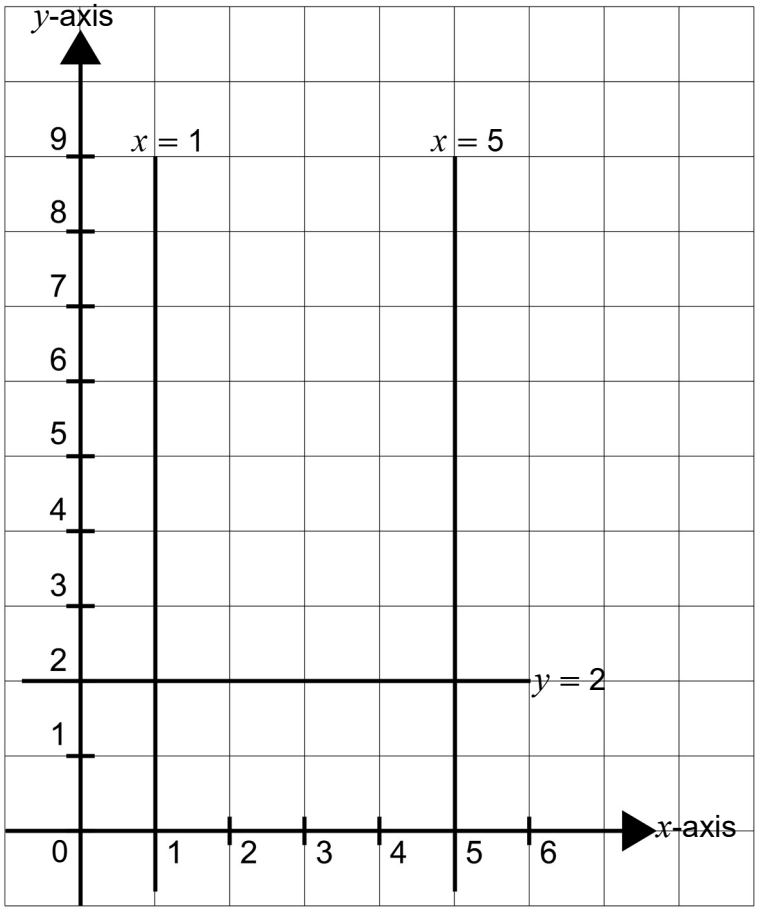
Marks

29. Three lines $x = 1$, $x = 5$ and $y = 2$ are shown on the graph below.



(a) On the same Cartesian plane, draw the graph of $y = x + 4$.

[2]

(b) What is the special name given to the figure enclosed by the four straight lines?

Answer: ..... [1]

Marks

(c) Calculate the **area** of the figure formed.

Answer: ..... square units [2]

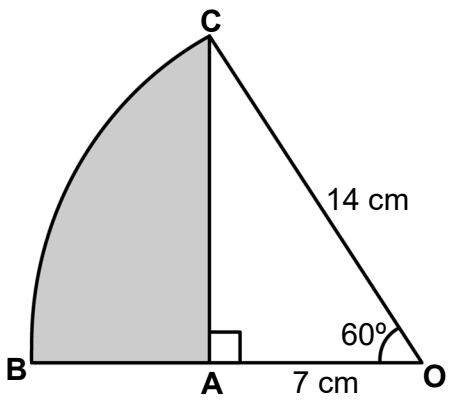
E

TL

CE

Marks

30. **OBC** is a sector of a circle with centre **O** and radius 14 cm. It is also given that **AO** = 7 cm.



[Given sin 60º = 0.87, cos 60º = 0.5, tan 60º = 1.73]

(a) Using one of the trigonometric ratios listed above, find the length of **AC**. Give your answer to the **nearest whole number**.

Answer: ..... cm [3]

(b) Find the length of arc **BC**. (Use $\pi = \frac{22}{7}$)

Answer: ..... cm [2]

Marks

(c) Using your answers to part (a) and part (b), find the **perimeter** of the shaded part **ABC**.

Answer: ……..……..….. cm [3]

E

TL

CE

Marks

31. A bag contains 12 red balls and x blue balls. A ball is chosen at random from the bag.

(a) (i) Write down in terms of x, the total number of balls in the bag.

Answer: ..... balls [1]

(ii) Find in terms of x, the probability that a **red** ball is chosen.

Answer: ..... [1]

(b) Given that the probability that a blue ball is chosen equals to $\frac{3}{5}$, find the **value** of x.

Answer: $x =$ ..... [3]

END OF PAPER

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